

FINANCIAL TRANSACTION ANALYSIS REPORT

Prepared by: FERDI KELANA PUTERA

Date: 24 June 2025

Data Loading & Initial Exploration

Objective:

Load the dataset and perform initial checks for structure, missing values, and basic statistics.

Choose Files

 No file chosen

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving test_1.csv to test_1.csv
user uploaded file "test_1.csv" with length 48731 bytes

Data Overview:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 5 columns):
Column Non-Null Count Dtype
--- --
0 Date 1000 non-null object
1 Description 1000 non-null object
2 Deposits 1000 non-null object
3 Withdrawls 1000 non-null object
4 Balance 1000 non-null object
dtypes: object(5)
memory usage: 39.2+ KB
None

First 5 Rows:

	Date	Description	Deposits	Withdrawls	Balance
0	20-Aug-2020	Cheque	3,391.02	00.00	83,839.30
1	20-Aug-2020	ATM	147,844.20	00.00	231,683.50
2	20-Aug-2020	IMPS	17,192.00	00.00	248,875.50
3	21-Aug-2020	Transfer	08.00	00.00	248,883.50
4	21-Aug-2020	Purchase	00.00	82,961.17	165,922.33

Descriptive Statistics:

	Date	Description	Deposits	Withdrawls	Balance
count	1000	1000	1000	1000	1000
unique	51	15	495	441	936
top	17-Sep-2020	Bill	00.00	00.00	00.00
freq	75	81	468	533	62

Data Cleaning & Transformation

Objective:

Convert monetary values from strings (e.g., "1,000.00") to numeric (float).
Parse Date into a datetime format for time-series analysis.

	Date	Deposits	Withdrawls	Balance
count	1000	1.000000e+03	1.000000e+03	1.000000e+03
mean	2020-09-15 01:12:00.0000000256	8.538883e+04	8.523763e+04	6.391821e+05
min	2020-08-20 00:00:00	0.000000e+00	0.000000e+00	0.000000e+00
25%	2020-09-04 00:00:00	0.000000e+00	0.000000e+00	9.426553e+03
50%	2020-09-16 00:00:00	4.000000e+00	0.000000e+00	2.510593e+05
75%	2020-09-27 00:00:00	2.006000e+03	5.195649e+04	9.416954e+05
max	2020-10-09 00:00:00	2.050674e+06	3.296138e+06	4.943629e+06
std	NaN	3.105901e+05	2.628950e+05	8.958291e+05

Key Financial Metrics

Objective:

Compute total deposits, withdrawals, and net cash flow.
Identify top transaction categories by frequency and value.

Financial Summary:
Total Deposits: \$85,388,826.40
Total Withdrawals: \$85,237,632.70
Net Cash Flow: \$151,193.70

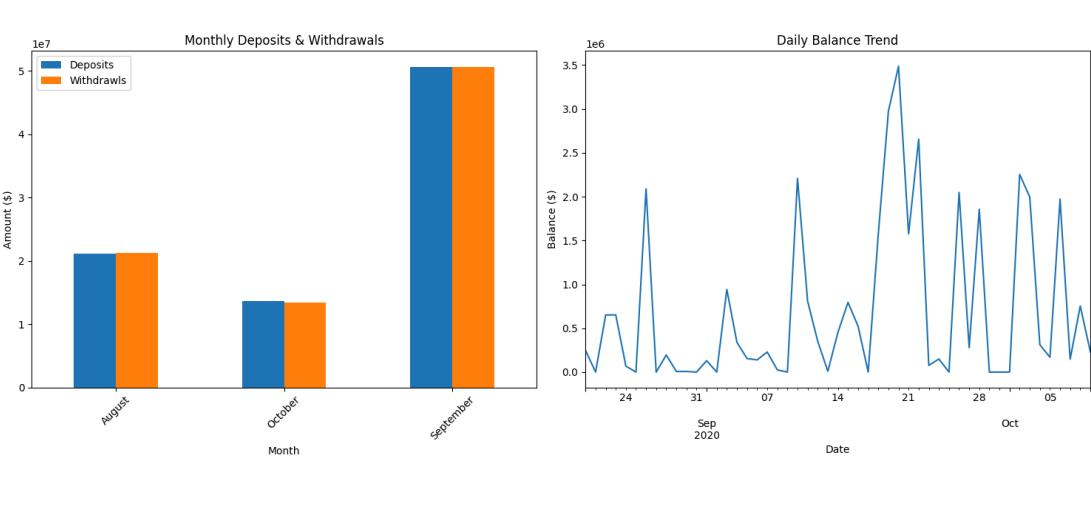
Top 5 Transaction Categories (Frequency):
Description
Bill 81
RTGS 76
Cheque 70
NEFT 70
Interest 69
Name: count, dtype: int64

Top 10 Categories by Transaction Value:
Description Deposits Withdrawls
NEFT 9818262.11 4447681.83
RTGS 9179956.53 6395981.93
Commission 8847851.67 9090169.63
Transfer 7843618.32 7167465.51
Cash 7365867.53 7739332.07
ATM 7022518.69 6811573.77
Cheque 5998191.84 7027293.98
IMPS 5613035.65 2018880.90
Reversal 4977259.12 8595384.28
Bill 4605713.25 4754444.62

Time-Series & Trend Analysis

Objective:

Analyze monthly trends in deposits/withdrawals.
Track daily balance changes over time.



Transaction Distribution & Outliers

Objective:

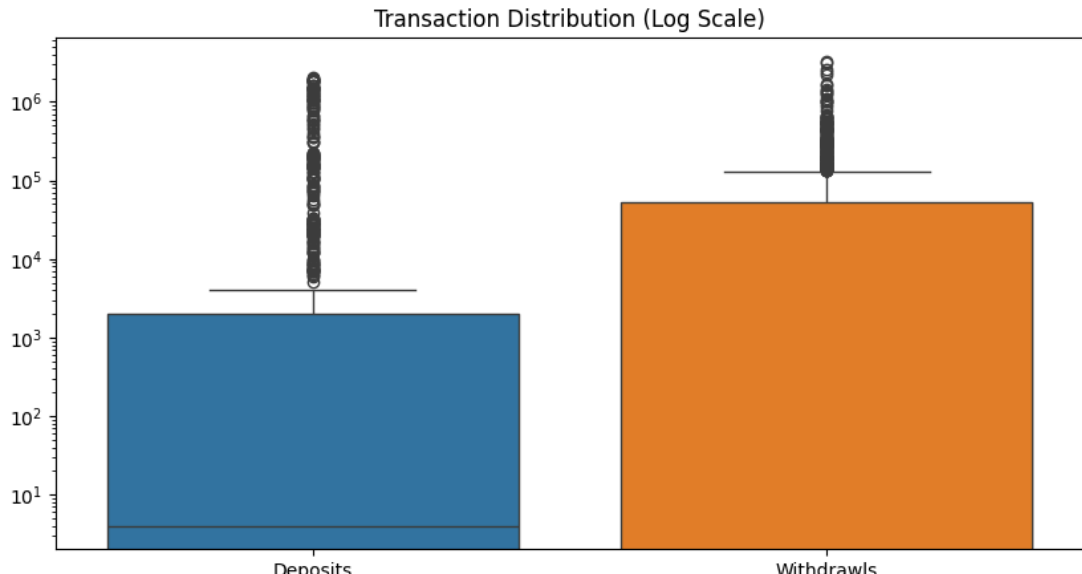
Identify outliers (unusually large transactions).
Analyze distribution of deposits/withdrawals.

Largest Deposits:

	Date	Description	Deposits
136	2020-08-26	NEFT	2050673.75
192	2020-08-31	NEFT	1994907.80
649	2020-09-22	NEFT	1933335.50
526	2020-09-17	RTGS	1914618.25
593	2020-09-18	ATM	1890558.00

Largest Withdrawals:

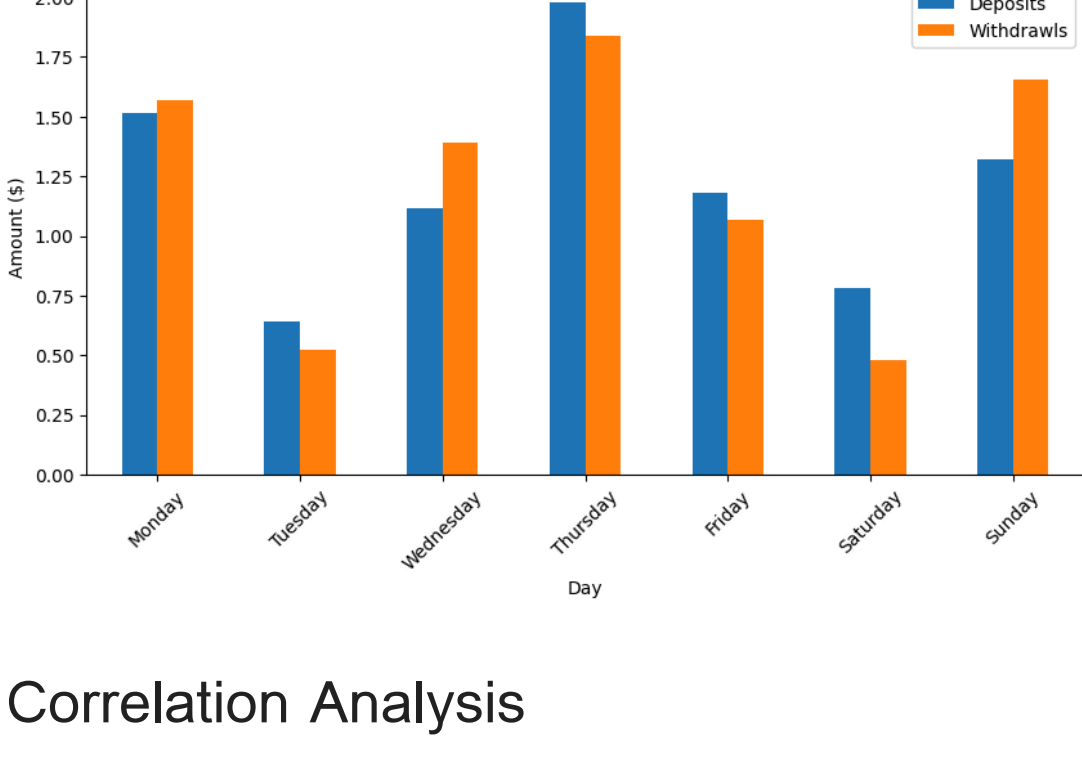
	Date	Description	Withdrawls
546	2020-09-17	Commission	3296138.38
936	2020-10-07	Cash	3113978.99
654	2020-09-23	RTGS	2559519.86
754	2020-09-27	Reversal	2382559.21
353	2020-09-11	Cheque	2210255.98



Day-of-Week Analysis

Objective:

Identify weekly patterns



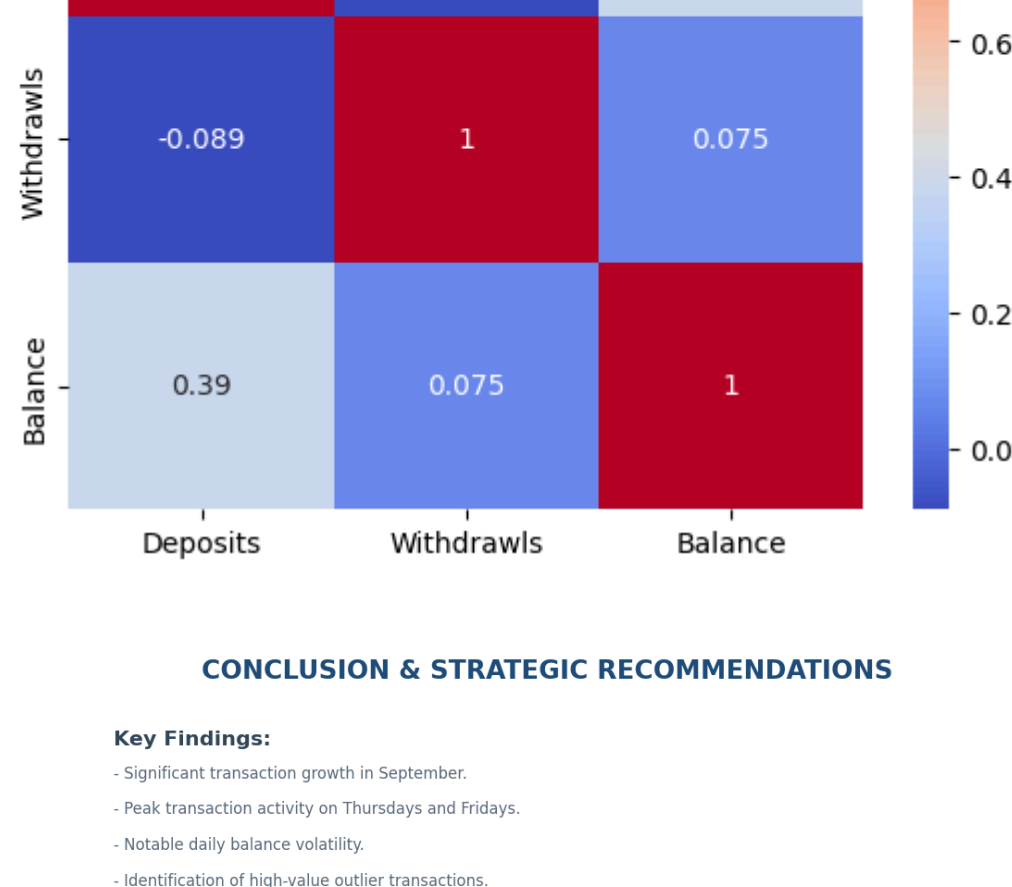
Correlation Analysis

Objective:

Check if deposits, withdrawals, and balance are correlated.

Correlation Matrix:

	Deposits	Withdrawls	Balance
Deposits	1.000000	-0.089227	0.387164
Withdrawls	-0.089227	1.000000	0.075243
Balance	0.387164	0.075243	1.000000



CONCLUSION & STRATEGIC RECOMMENDATIONS

Key Findings:

- Significant transaction activity growth in September.
- Peak transaction activity on Thursdays and Fridays.
- Notable daily balance volatility.
- Identification of high-value outlier transactions.

Actionable Recommendations:

- Implement enhanced cash flow forecasting based on weekly patterns.
- Develop an alert system for large transactions to streamline review.
- Optimize operational staffing for peak days (Thurs/Fri).
- Investigate drivers of September's transaction surge for future planning.

This analysis aims to empower data-driven decisions.
We welcome further discussion on these insights.